Learn From Past to Future: Exploiting Self-Training and Curriculum Learning in Remote Sensing Class-Incremental Semantic Segmentation

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***Abstract*—Class-incremental semantic segmentation focuses on updating the segmentation model with only new-class samples. Catastrophic forgetting and background shift are the two prevalent challenges. We identify two additional issues in remote sensing data that worsen these problems: significant class distribution variability and error accumulation-induced model degradation. To solve these three problems, we propose a new self-training and curriculum learning guided dynamic refined network (STCL-DRNet). First, we introduce a self-training auxiliary branch to complement the frozen last-step model, integrating cross-step knowledge to mitigate rapid forgetting. Then, a gradient-oriented dynamic refined loss is proposed to assess under-learned classes and mitigate class imbalance. Furthermore, class-balanced curriculum learning (CL) is embedded to alleviate performance degradation throughout incremental training. Extensive experiments on benchmark datasets, including DeepGlobe, iSAID, ISPRS Potsdam, and Vaihingen, demonstrate that the proposed STCL-DRNet achieves state-of-the-art (SOTA) performance. In the 1-1s setting of the DeepGlobe dataset, STCL-DRNet exceeds previous SOTA methods by 11.6% in mIoU. For the iSAID 10-1s setting, it outperforms the previous SOTA by 12.76% in mIoU. As for ISPRS Potsdam and Vaihingen, our STCL-DRNet surpasses the SOTA by 5%–8% in all settings. Visualization and analysis further validate its interpretability. The our code is available at https://github.com/cv516Buaa/STCL-DRNet**

***Index Terms*—Class-incremental semantic segmentation, curriculum learning (CL), remote sensing, self-training.**

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I. Introduction

THE remote sensing technology has been widely applied to various real-world vision tasks, such as remote sensing scene classification, object detection, and semantic segmentation. Among these tasks, remote sensing image semantic segmentation plays an important role on recognizing a greater variety of geographical elements at the pixel level. The advancements of deep convolutional neural networks have led to remarkable achievements in semantic segmentation tasks. Various semantic segmentation models based on deep learning methods [1], [2], [3], [4], [5], [6] have shown excellent performance. Most methods employ an encoder–decoder model structure to extract semantic features and generate pixel-level labels, such as Unet [2], DeeplabV3 [7], DAnet [4], Segformer [5], and Segmenter [6]. All of these methods assume a common training protocol in which training samples of all categories remain static and available throughout the training process.

However, many real-world applications require models to adapt to an evolving sequence of training samples with novel classes. To satisfy this dynamic training process, a class-incremental semantic segmentation task has been proposed. In the process of class-incremental training, most models will suffer severe forgetting of the old classes which is known as catastrophic forgetting [8]. The absence of annotations for learned classes in data sequences undermines the model's capability to distinguish between background elements and previously learned classes. This phenomenon is also referred as background shift [9].

To alleviate the above problems, several methods have been proposed to tackle class-incremental semantic segmentation tasks in both natural scenes [9], [10], [11], [12] and remote sensing scenes [13], [14], [15], [16], [17]. Most existing class-incremental methods can be generally divided into two categories: regularization-based and replay-based methods. Regularization methods [9], [10], [11], [13], [14], [15], [17] focus on imposing constraints on the optimization objectives to mimic the output of the previous model with respect to the old classes, such as knowledge distillation and pseudo labels. Replay-based methods [11], [12], [16] are dedicated to enriching old-class training data by storing previous samples [11], collecting web-crawled images [12], or designing generative networks [12], [16].

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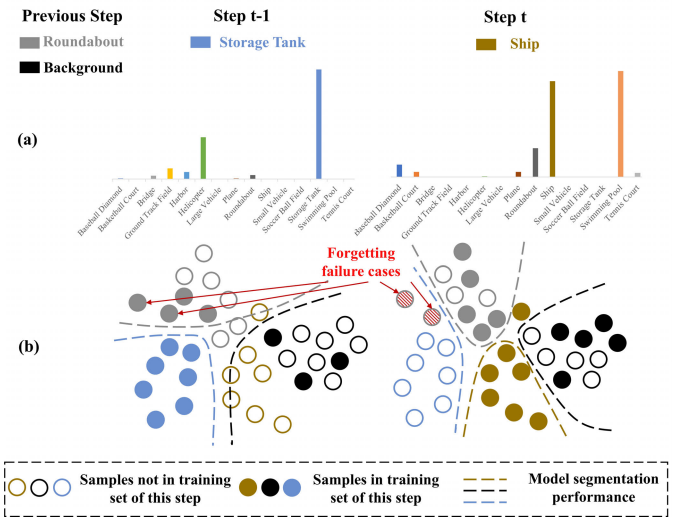


Fig. 1. Illustration of the challenges in incremental semantic segmentation for remote sensing scenes, using the iSAID dataset as an example. We assume that the model has already learned two classes: *background* and *roundabout*. In the next two steps, the model learns two additional classes: *storage tank* and *ship*. (a) Shows the pixel class distribution difference in images for the two classes. (b) Illustrates how the training target can shift across steps due to changes in the training samples. These two issues accelerate forgetting.

Due to the high resolution of remote sensing images and the rich variety of geographical objects, regularization methods mainly have the following three problems.

1) Catastrophic Forgetting and Background Shift: These are common challenges in class-incremental learning, which are exacerbated in remote sensing by the following issues.

2) Unbalanced and Evolving Class Distribution: Geographical objects in aerial images often vary greatly in scale and context. As shown in Fig. 1(a), the class distribution and interclass correlation in the iSAID dataset differ significantly and change across training steps. The distribution and contextual variations cause the model to overfit to newly introduced classes.

3) Model Degradation From Absent Old Classes and Error Accumulation: In regularization-based approaches, the teacher model often produces erroneous pseudo labels. These errors propagate through subsequent training steps, degrading the model's performance. Moreover, as shown in Fig. 1(b), the absence of some samples from old classes like *roundabout (gray)* and *storage tank (blue)* lead to overfitting on incorrect targets based on limited data.

To address these problems, we propose the self-training and curriculum learning (CL) guided dynamic refined network (STCL-DRNet). Fig. 2 reveals the paradigm of our proposed STCL-DRNet in contrast to previous methods. First, to mitigate catastrophic forgetting, we introduce a dual-teacher framework. Alongside the standard frozen teacher from the previous step, a slowly evolving teacher is maintained via exponential moving average (EMA). The EMA teacher has the potential to stably integrate knowledge across old and new classes. Second, to handle the uneven class distribution, we design a dynamic refined loss. This loss calculates a gradient-oriented under-learning score for each class, dynamically assigning greater weight to harder classes. Third, to reduce noise from pseudo labels and prevent model degradation, we implement a class-balanced CL strategy. This strategy filters pseudo labels based on their posterior probability, adopting an easy-to-hard training schedule to minimize the impact of low-quality supervision. Together, these components enable robust incremental learning.

This article is extended from our conference paper [18].[1] The extensions are summarized as follows.

1) We make a detailed mathematical analysis and explanation of EMA teacher in Section III-C, dynamic CL in Section III-D, and dynamic refined loss in Section III-E. More key derivations and implementation rationales are exposed.

2) To further validate the performance of our proposed method, we additionally present the experimental results on two other semantic segmentation datasets in remote sensing scenes: ISPRS Potsdam and Vaihingen.

3) We supplement more ablation experiments and visualization results to demonstrate the effectiveness and our method.

Following [14], [16], and [17], we benchmark our unified STCL-DRNet framework on four remote sensing datasets [19], [20], [21]. Encouraging results demonstrate the effectiveness of STCL-DRNet, achieving state-of-the-art (SOTA) performance in the vast majority of settings and classes. Our main contributions are summarized as follows.

1) A dual-teacher framework that combines a frozen prior-step model with an EMA-based self-training teacher to mitigate catastrophic forgetting and background shift.

2) A gradient-oriented dynamic refined loss that automatically reweights classes based on their learning difficulty in uneven class distribution.

3) A class-balanced CL strategy for pseudo-label generation, which adopts an "easy-to-hard" training manner to reduce noise and prevent model degradation.

4) The unified framework STCL-DRNet is evaluated on four benchmark datasets: DeepGlobe [19], iSAID [20], ISPRS Potsdam, and Vaihingen [21]. It achieves SOTA results across 12 incremental settings in total.

The article is structured as follows. Section II reviews recent relevant studies and their connections to our method. Section III details the technical details and implementation of our method. Section IV presents experimental results, including performance comparisons, visualization outcomes, and in-depth discussions. Finally, Section V summarizes the main contributions.

II. Related Works

A. Semantic Segmentation

As a basic task in computer vision, semantic segmentation aims at assigning a label for every pixel in an image and has achieved great success due to the development of the deep neural network. Fully connected network [1] and Unet [2], as the first end-to-end models for semantic segmentation, only use the convolution layers to implement pixel-by-pixel segmentation. Based on the FCN [1] architecture, some methods [7], [22] are designed for increasing the receptive field to exploit contextual information. In instance, DeepLabV3 [7] applies different-rate atrous convolution in order to assemble multiscale information. Some works introduce the attention

[1]IEEE International Symposium on Geoscience and Remote Sensing (IGARSS) oral paper.

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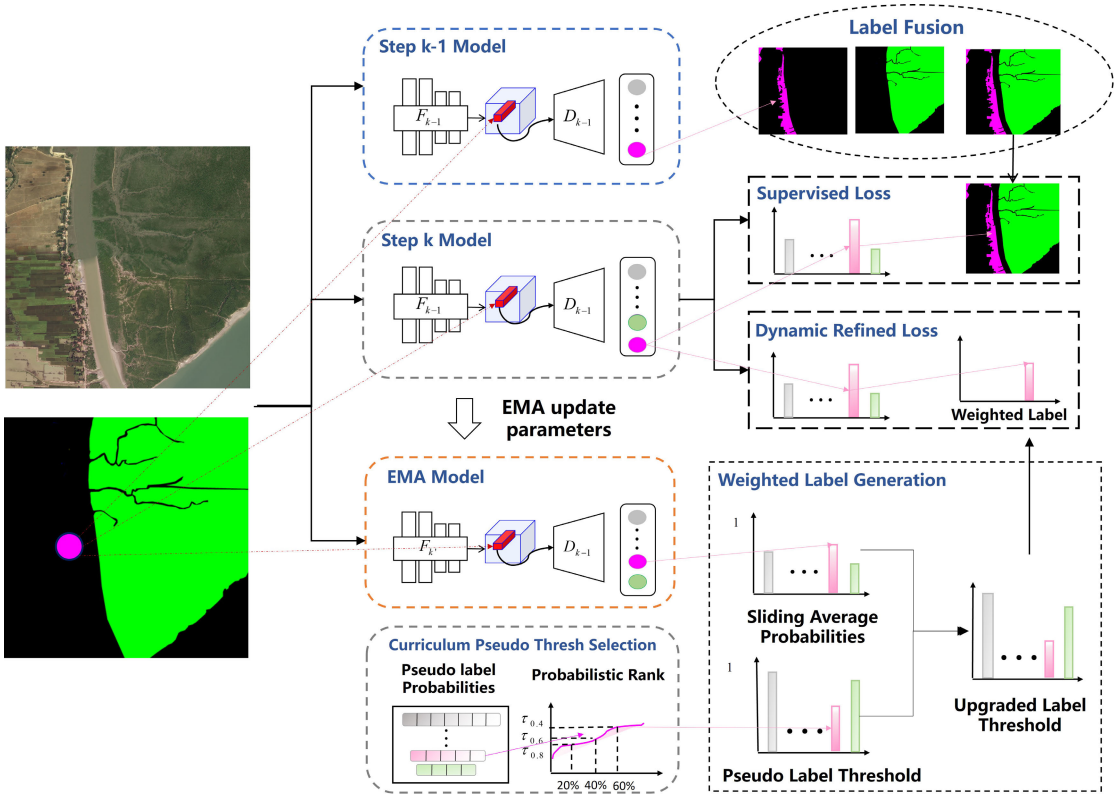


Fig. 2. Overall framework of our proposed STCL-DRNet. Specifically, images are simultaneously fed into the current step training model, the frozen last-step model, and the self-training EMA model during the training process. Pseudo labels for old classes are generated and integrated with current labels to generate supervision loss. EMA model updated with the EMA of current step model parameters employs a class-balanced CL strategy to gradually generate "easy-to-hard" pseudo labels and dynamic refined loss. These two losses jointly guide the optimization of STCL-DRNet.

mechanism [3], [4], [23], [24] to convolution networks for better image understanding, such as DANet [4] and PSANet [3]. Transformer architectures [5], [6], [25], [26] have been applied in semantic segmentation since the vision transformer was proposed in the vision recognition. Although these methods show great effectiveness on the semantic segmentation task, there are still some shortcomings faced with the realistic application, where the new classes training data was incrementally exposed to the models. Severe reduction is found in the model's performance which is called catastrophic forgetting. Our work focuses on mitigating catastrophic forgetting and achieve the class-incremental learning in semantic segmentation models.

B. Class-Incremental Learning

The incremental learning methods have been studied in the past years since the catastrophic forgetting was first proposed in the research of connectionist networks [8]. Catastrophic forgetting means a model will suffer great losses on the ability of old tasks when it is trained on a data stream with sequentially added new tasks data. For class-incremental learning, the tasks can be different-class image recognition. The class-incremental learning for image classification has been commonly researched and can be divided into three categories: rehearsal-based methods, architectural methods, and regularization-based methods. The rehearsal-based methods usually store a few samples of the raw images [27], [28], [29] or the embedding of the intermediate features [30], [31] to help the model recall old-class knowledge. Some of the rehearsal methods use the generative network [32], [33], [34] to produce the old-class images or features. Architectural methods [35], [36] dynamically expand new branches of the network or mask parts of the network and parameters for different tasks. Regularization methods assume that the old-class data is not available but the model trained on the old-class data is available, which can be divided into parameter punishment methods and distillation-based methods. Parameter punishments methods [37] apply punishment on important parameters judging to prevent their changes. Distillation-based methods [38], [39] keep the output of the current training model mimic the old model by distilling their output logits or the intermediate features.

Compared to the development of class-incremental classification, class-incremental semantic segmentation has attracted attention in recent few years. As a dense prediction task, class-incremental semantic segmentation methods are designed to deal with not only catastrophic forgetting but background shift, which refers to the semantic change of background pixels as some of them are marked as new classes in the following training steps. The current incremental semantic segmentation methods can be divided into regularization-based method [9], [10], [11], [12], [13], [14], [15], [17], [40], replay-based methods [16], [41], [42], [43]. Regular-based methods utilize distillation or pseudo labels from the old

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trained model to preserve the knowledge. MiB [9] applies models the background class probability for the distillation between the past step model and current step model. PLOP [10] utilizes the past step model to generate pseudo labels and uses distillation with multiscale pooling of the intermediate features. As for the replay-based method, for example, SSUL [11] stores a fixed amount of old classes' images and labels to replay in the following steps. RECALL [12] uses web-crawled images or a GAN-based network to generate images as a supplement.

In the field of remote sensing, the methods mentioned above, like KD methods, pseudo-label generating, and replay methods have proven the effectiveness. Shan et al. [14] distilled the multilevel features from different layers by calculating similarities between two models to help the inheritance of capabilities of the old model. Li et al. [13] designed diversity distillation loss and a multihead classifier fusion mechanism to strengthen the small object learning. Rong et al. [15] proposed a historical information-guided feature distillation to obtain the global information for foreground old classes. Shan et al. [16] designed a generating network to produce intermediate features for distillation to keep memories for the old class. Rong et al. [17] model cross-image relationship for class-incremental semantic segmentation to cope with the continual learning problems in the feature space. Different from the above methods, our work introduces a new multiteacher incremental segmentation framework, combined with CL and self-training.

C. CL and Self-Training

CL is a training strategy which mimics human learning manners from easy to hard. From the perspective of optimization, it gradually optimizes the problem from a smooth version to a less smooth version and is believed to help the optimization achieve the better local minima [44]. Furthermore, some researches [44], [45] propose that the process from easier samples to harder samples is beneficial to denoising the noisy data for harder samples. Considering the motivation of denoising, CL is utilized in domain adaptation and weakly supervised, semi-supervised, or unsupervised learning. For example, Gong et al. [46] designed a multimodal CL strategy to investigate the difficulty of classifying every unlabeled image and integrate the information from different feature modalities for semi-supervised image classification. For semantic segmentation, CL has been introduced in domain adaptation and semi-supervised semantic segmentation. In instance, Feng et al. [47] proposed a class-balanced CL method to exploit the noisy pseudo labels for semi-supervised semantic segmentation. Similarly, the class-incremental learning process can be regarded as a semi-supervised learning for the past classes' data, given that the labels for the learned classes are inaccessible. Our work is the first method to utilize CL in the generation of the old classes' pseudo labels to get better performance in the class-incremental semantic segmentation.

Self-training methods [48], [49], [50], [51], [52] aim at utilizing unlabeled images in the semi-supervised training process by generating high-confidence pseudo labels for unlabeled images. These methods usually train a model as the teacher model based on the labeled data and then use the teacher model to generate pseudo labels. However pseudo labels might be over-confident and different methods [48], [49], [50], [51], [52] are proposed to deal with the noisy pseudo labels. For class-incremental semantic segmentation, the old classes' pixels are unlabeled, and the trained model is used as a teacher model to generate pseudo labels. As a result, the noise in the pseudo label causes the severe forgetting of old classes. In order to solve the noisy pseudo-labels problem, our work designed a self-training method combined with a CL strategy and self-training methods.

III. Methodology

A. Problem Definition

Class-incremental semantic segmentation aims to train a segmentation model in T steps and gradually acquire segmentation capabilities of more and more classes in sequential training sets with less forgetting. For each step t ($t \in 1, 2, 3 \ldots, T$), dataset D^t is accessible which is composed of image and label pairs (X^t, Y^t). $X^t \in R^{H\times W\times 3}$ denotes an input image and $Y^t \in C^{t,H\times W}$ denotes the corresponding ground truth segmentation mask, where H and W denote the image height and width, respectively. For the annotated classes of training set labels Y^t, $C^t = \{c_i\}_{i=1}^{|C^t|}$ is the class set in the current step dataset D^t while the class set in different steps are distinct $C^i \cap C^j = \emptyset$ $(i \neq j)$ excluding background class c_b. Moreover, old classes $C^{0:t-1}$ and future novel classes $C^{t+1:T}$ pixels in current step label set Y^t are annotated as c^b. The class-incremental semantic segmentation assumes that in the new step t the learned dataset $\{D^1, D^2, \ldots, D^{t-1}\}$ are invisible to the current training model. Consequently, the model f^t is finetuned from the training set D^t and the last-step model f^{t-1} at the tth step. It is expected to acquire the segmentation capability of all learned classes $C^{0:t} = \{C^0, C^1, \ldots, C^t\}$ of all training steps.

Typically, a semantic segmentation model f of incremental step t is composed of an encoder F^t for feature extraction and a decoder G^t for generating a prediction. The encoder can extract high-level features through cascade network layers $Z^t = F^t(X^t)$. The features are fed into the decoder G^t to produce the logit map $S^t = f^t(X^t) = G^t(Z^t) = G^t \circ F^t(X^t)$ and generate class probabilities of each pixel $P^t = \text{Softmax}(S^t)$. The corresponding learnable parameters for step t models can be denoted as $\Theta^t = \{\theta_l^t \ldots \theta_l^t\}$ with $l \in \{1 \ldots L\}$ denoted as the layer number and the step t model is denoted as f^t.

B. Overall Framework

In general, for most regularization-based methods of class-incremental semantic segmentation, the overall framework is depicted as Fig. 3(a). For step 0, the segmentation model is typically trained from scratch, following the traditional training process by generating supervised cross-entropy loss L_{ce}. Nonetheless, in step t ($t > 0$), the last-step model f^{t-1} with frozen parameters Θ^{t-1} is employed to provide old-class knowledge for the current model f^t by generating the pseudo labels or distillation from Intermediate output features. Apart from the supervised loss L_{ce}, the regularization loss L_{reg} will

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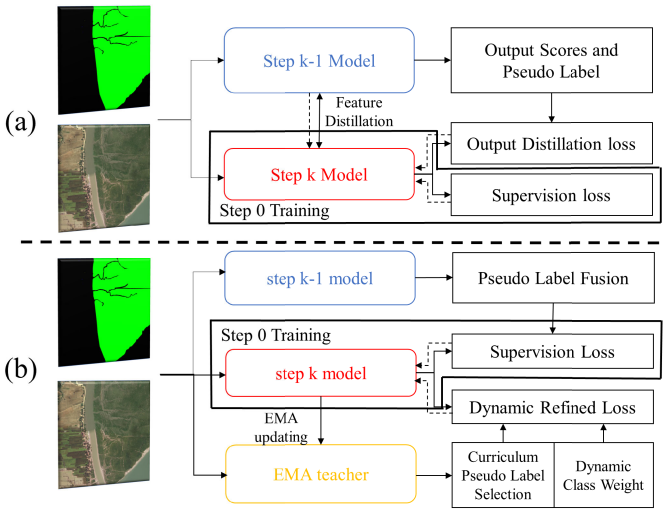


Fig. 3. Difference between most regularization-based methods and ours is illustrated in (a) and (b). In previous methods, the last-step frozen model is typically used as a teacher model to distill old-class information either through knowledge distillation from the intermediate layers or the output layers, along with pseudo labels. In contrast, our method introduces an EMA self-training copy as an additional teacher model, apart from the frozen last-step model, with pseudo-label fusion. We employ a CL strategy to select the appropriate samples and compute dynamic refined loss.

be computed to optimize the preservation of knowledge from previously learned classes.

By contrast, our STCL-DRNet framework is shown as Fig. 3(b). Contrary to the regularization-based methods framework, we introduce a self-training branch to serve as another teacher model for better integration of knowledge from past and current classes. On the basis of EMA teacher output, a CL strategy in the pseudo-label selection is used to mitigate noisy, erroneous output impact, especially in multistep incremental learning. Considering class distribution imbalance in remote sensing scenes, we design a dynamic refined loss L_{dr} to copy with weak-class forgetting problems. Furthermore, we aggregate old-class knowledge distillation and new-class learning by merging the pseudo labels from model f^{t-1} and labels Y^t for simplicity in model optimization. The pseudo possibility P^{t-1} for each pixel i and old class $c \in C^{1:t-1}$ on image X^t is computed as (1). The corresponding pseudo label for old classes is computed as (2)

$$\widetilde{P}_i^{t-1}(c) = \mathrm{Softmax}\left(f^{t-1}\left(X^t\right)[h_i, w_i, c]\right) \tag{1}$$

$$\widetilde{Y}_i^{t-1} = \underset{c \in C^{0:t-1}}{\arg\max}\left(P_i^{t-1}(c)\right) \tag{2}$$

where (h_i, w_i) denotes horizontal and vertical coordinates of the pixel i. Softmax denotes the softmax computation function to converts a vector into a probability distribution.

A pseudo-threshold τ is roughly set to assess the confidence of the generated pseudo labels. The labels with maximum pseudo possibility lower than τ will be classified as c_b (background). We use the generated pseudo labels, fused with ground truth labels, to compute the supervision loss L_{ce}.The process of label fusion and modified loss L_{ce} is formulated as (3) and (4)

$$\hat{Y}_i^t = \begin{cases} Y_i^t & \text{if } Y_i^t \neq c_b \\ c & \text{if } Y_i^t = c_b \text{and } \widetilde{P}_i^{t-1}(c) > \tau \\ & \text{and } \widetilde{Y}_i^{t-1} = c, c \in C^{0:t-1} \\ c_b & \text{otherwise.} \end{cases} \tag{3}$$

$$L_{\mathrm{ce}} = -\frac{1}{HW}\sum_{i=1}^{HW} \log P_i^t\left(\hat{Y}_i^t\right). \tag{4}$$

Current step model f^{t-1} is optimized simultaneously under L_{ce} from the fused pseudo labels and L_{dr} from the EMA model with CL strategy. The overall loss was shown as follows:

$$L = L_{\mathrm{ce}} + \alpha L_{\mathrm{dr}}. \tag{5}$$

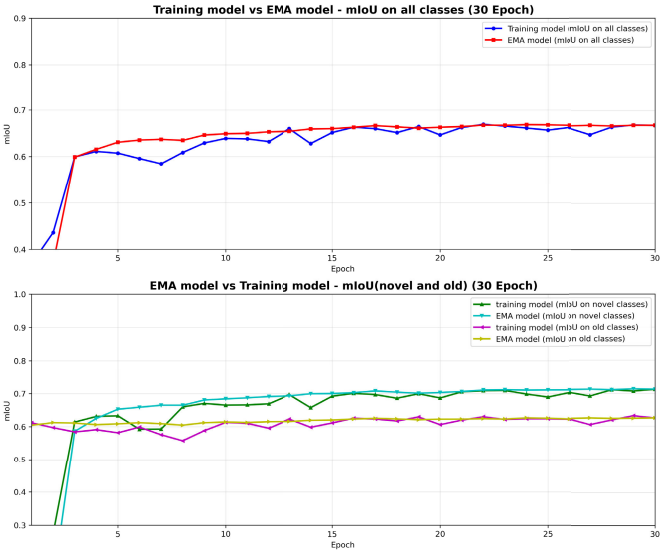


Fig. 4. Performance of training model versus EMA model: mIoU on all, old, and new classes (DeepGlobe 3-3, 30 epochs).

C. EMA Self-Training Teacher

Self-training methods are widely employed in knowledge transfer tasks, such as domain adaptation and semi-supervised learning, where models are trained to adapt to datasets with differing distributions. For class-incremental semantic segmentation, datasets D_t in the current step usually contains the learned classes without separate annotations, but only background c_b. Learning from these unlabeled old-class data with the last-step model f_{t-1} can be considered a form of semi-supervised learning on previously known classes. However, there is a distribution shift of images among different steps. For the old classes in $C^{0:t-1}$, the differing class distribution results in noisy outputs from f^{t-1}, leading to rapid overfitting of the current model f^t on the new step images. Classes that experience significant distribution shifts are more prone to being forgotten.

As shown in Fig. 4, we evaluate the performance of both the original and the EMA model on old and new classes in step 1 of the DeepGlobe 3-3 setting. The EMA-updated model exhibits superior segmentation stability compared to the training model in the early stages. Notably, for old classes,

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the EMA-updated model does not exhibit the significant performance drop seen in the standard training model during the early phase. Concurrently, it maintains a competitive ability to learn new classes, resulting in stable and balanced performance in incremental training.

To better use the unlabeled old classes, we introduce a self-training teacher branch $\overline{f}$ which parameters $\overline{\Theta}$ is updated with an EMA of the current model f^t parameters weights Θ_t. For the iteration m in the training step t, the updating process is shown as follows:

$$\overline{\Theta}_m = \gamma_m \overline{\Theta}_{m-1} + (1-\gamma_m)\,\Theta_m^t \tag{6}$$

where γ_m denotes the decay rate which restricts the contribution of the current model's weight after iteration m. The EMA-updated branch $\overline{f}$ with parameters $\overline{\Theta}_m$ can be regarded as a more gradually and subtly optimized version of the current model f^t. The characteristic is evident from the derivation process of the EMA algorithm as follows:

$$\begin{aligned}
\overline{\Theta}_m &= \gamma_m \overline{\Theta}_{m-1} + (1-\gamma_m)\,\Theta_m^t \\
&= \gamma_m \gamma_{m-1} \overline{\Theta}_{m-2} + \gamma_m (1-\gamma_{m-1})\,\Theta_{m-1}^t + (1-\gamma_m)\,\Theta_m^t \\
&= \prod_{k=1}^{m} \gamma_k \cdot \Theta_1^t + \sum_{j=1}^{m} \left((1-\gamma_j) \cdot \prod_{k=j+1}^{m} \gamma_k \cdot \Theta_j^t \right) \\
&= \prod_{k=1}^{m} \gamma_k \cdot \Theta_1^t + \sum_{j=1}^{m} \left((1-\gamma_j) \cdot \prod_{k=j+1}^{m} \gamma_k \cdot \left(\Theta_1^t - \sum_{k=1}^{j} \nabla_L \left(\Theta_k^t \right) \right) \right) \\
&= \Theta_1^t - \sum_{k=1}^{m} \left(1 - \prod_{j=k}^{m} \gamma_j \right) \nabla_L \left(\Theta_k^t \right).
\end{aligned} \tag{7}$$

As shown in (7), the gradient in each Iteration is multiplied by a coefficient $1 - \prod_{j=k}^{m} \gamma_j$ which is much lower than 1 initially. It will reduce the influence of recent gradients on the current parameters. The coefficient with respect to the previous gradient will gradually increase to 1 in the following iterations. The parameters of the EMA-Teacher model are less influenced by the immediate gradients. This makes it less prone to rapidly fitting the new data distribution, potentially allowing it to retain more knowledge from previous training steps and keep a better balance. We utilize this slower EMA branch as an additional teacher to mitigate the forgetting of old knowledge due to distribution shifts. The stability of the EMA model makes it well-suited for CL, enabling reliable noise reduction and robust outputs. The outputs from the EMA branch are combined with our dynamic CL to compute the Dynamic Refined Loss L_{dr} as a supplement and amendment for loss L_{ce}. The methods will be detailed in Sections III-D and III-E.

D. Dynamic CL

The EMA model provides a model branch that ensembles multiple versions throughout the training process. Since the segmentation output of the model cannot be entirely accurate, pseudo-label generation inevitably introduces noise, compounded by the presence of distribution shift. Learning from noisy model outputs can degrade segmentation performance, particularly for underrepresented or rare classes. Moreover, in multistep incremental training, this issue exacerbates catastrophic forgetting. To address the challenge of noisy outputs, we propose a dynamic CL strategy that selects pseudo labels in an easy-to-hard manner.

Algorithm 1 Algorithm of Dynamic CL

Input: Training images and labels $(X^t, Y^t) \in D^t$, EMA updated model $\overline{f}$, step t training model f^t
Output: Best training model for current step f^t

```
Initialize f^t and f̄ from last step model f^{t-1}
Divide the total training epochs into S round
for each s ∈ [1, 2, 3, 4, ..., S] do
    Initialize queue̅_c = [] for c ∈ C^{0:t}
    Compute the ratio γ_s = s/S
    for each X^t ∈ D^t do
        Inference f̄ on X^t ⟶ P̄^t and Ȳ^t
        for each i ∈ [1, 2, ..., HW] do
            queue̅_c = queue̅_c append(P̄_i) if Ȳ_i = c
        end for
    end for
    τ_{c,s} = descend-sort(queue̅_c)[γ_s · length(queue̅_c)]
    for each epoch ∈ epoch_s do
        for each X^t ∈ D^t do
            Inference f^{t-1} on X^t ⟶ Ŷ^t
            Inference f̄ on X^t with τ_{c,s}, G_c ⟶ Ȳ^t
            Optimize model f^t on (X^t, Ŷ^t, Ȳ^t) by loss L
            if iterations%ζ is 0 then
                Update the EMA model f̄ and G_c
            end if
        end for
    end for
end for
return: f^t
```

The dynamic CL strategy can be summarized as Algorithm 1. During step t of incremental learning, the training process is divided into S stages, with an increasing ratio β_s assigned to each stage s. At the beginning of each stage s, the EMA model performs inference on the entire dataset D^t to gather class-wise pixel-level probabilities to form a queue $\overline{\text{queue}}_c$ for each old class c in $C^{0:t-1}$. After sampling and sorting the collected pixel-level output probabilities in $\overline{\text{queue}}_c$, the top γ_s ratio of values are selected as the threshold $\tau_{c,s}$ for pseudo-label generation of class c in stage s. The whole process can be denoted as follows:

$$\overline{P_i^t} = \max_{c \in C^{0:t-1}} \left(\text{Softmax} \left(\overline{f} \left(X^t \right) [h_i, w_i, c] \right) \right) \tag{8}$$

$$\overline{Y_i^t} = \operatorname*{argmax}_{c \in C^{0:t-1}} \left(\text{Softmax} \left(\overline{f} \left(X^t \right) [h_i, w_i, c] \right) \right) \tag{9}$$

$$\overline{\text{queue}}_c = \overline{\text{queue}}_c \text{ append } \left(\overline{P_i} \right) \text{ if } \overline{Y_i} = c \tag{10}$$

$$\tau_{c,s} = \text{descend-sort } \left(\overline{\text{queue}}_c \right) \left[\beta_s \cdot \text{length } \left(\overline{\text{queue}}_c \right) \right] \tag{11}$$

where X^t denotes the input image in current training dataset D^t and $i \in 1, 2 \ldots H \cdot W$ the pixel index in image X^t. append refers to adding elements to the end of an array, and descent-sort is the descending sorting operation of an array. The threshold $\tau_{c,s}$ serves as the criterion to determine whether

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the pixel-wise pseudo label is easy enough in the CL stage s. Pseudo labels with probabilities below the threshold $\tau_{c,s}$ are ignored. At the first stage, we feed the model with easy pseudo labels with probabilities exceeding than $\tau_{c,1}$. In the subsequent curriculum training stages, we gradually increase γ_s to sample harder pseudo labels with softmax probability higher than the decreasing $\tau_{c,s}$ from the EMA branch. The $\overline{\text{queue}_c}$ will be cleared at the end of each learning stage. Considering the presence of weak or rare classes, our CL strategy incorporates a bias during pseudo-label selection to prioritize these classes. In Section III-E, we propose a dynamic loss whose computation utilizes the CL threshold $\tau_{c,s}$ and dynamically decays this threshold based on the learning progress of different classes.

E. Dynamic Refined Loss

For underrepresented or rare classes, insufficient learning supplementation is a common issue. The outputs for these classes from the last-step model f^{t-1} are noisier and more erroneous. The learning imbalance exacerbates varying paces of catastrophic forgetting especially in multisteps incremental learning. To overcome this issue, we propose a dynamic refined loss using $\tau_{c,s}$ from dynamic CL and outputs from the self-training model $\overline{f}$. To compensate for under-learning in weak classes, we adopt a gradient-oriented assessment to compute the under-learning scores for each class c in the currently training model f^t. For pixel i of image X^t, the gradient g_i of L_{ce} with respect to the output of the ith neuron in the final output layer is as follows:

$$g_i = \left|\frac{\partial L_{\text{ce}}}{\partial Z_i}\right| = \left|\widetilde{P}_i^t\left(\widetilde{Y}_i^t\right) - 1\right| = 1 - \widetilde{P}_i^t\left(\widetilde{Y}_i^t\right) \tag{12}$$

where $Z_i = f^t(X^t)(\widetilde{Y}_i^t, i)$ denotes the last layer output of pixel ith and $\widetilde{Y}_i^t$th channel of image X^t. $\widetilde{Y}_i^t$ is the pseudo label of pixel i of model f^t.

For each class c, we maintain momentum-average gradient scores $\mathcal{G}_c$ updated using an EMA of scores g_i. The computing process is denoted as follows:

$$\mathcal{G}_c = \frac{\gamma_m \cdot g_i \cdot \mathbb{I}\left(\widetilde{Y}_i^t == c\right)}{\left\|\mathbb{I}\left(\widetilde{Y}_i^t == c\right)\right\|} + (1 - \gamma_m) \cdot \mathcal{G}_c \tag{13}$$

where $\mathbb{I}()$ denotes the indicator function which is 1 when the condition is satisfied. $\|\ \|$ represents the norm function to count the pixels which pseudo labels are c. The momentum average score $\mathcal{G}_c$ is updated in the whole training step t and utilized to monitor the under-learning status of each class in real-time. For the pseudo-label selection of EMA model $\overline{f}$, we use the $\mathcal{G}_c$ to dynamically decay the threshold $\tau_{c,s}$ and generate pseudo labels $\overline{Y}_i^t$ as follows:

$$\overline{\tau}_{c,s} = (1 - \mathcal{G}_c) \cdot \tau_{c,s} \tag{14}$$

$$\overline{\mathcal{Y}}_i^t = \begin{cases} c & \text{if } \overline{Y}_i^t = c \text{and } \overline{P}_i^t(c) > \overline{\tau}_{c,s} \\ & c \in C^{0:t-1} \\ \text{IGNORE} & \text{otherwise.} \end{cases} \tag{15}$$

As evident in (12)–(15), we set a lower pseudo-label threshold to sample more pixels for classes with higher gradient-oriented under-learning scores. As a result, our method dynamically controls pseudo-label selection of model $\overline{f}$ based on predictions from the current training model f^t, thereby facilitating the selection of additional potential samples for weak classes. In addition, we design a dynamic refined loss to assign different weights for every class c according to its under-learning score $\mathcal{G}_c$. It can be denoted as follows:

$$L_{\text{dr}} = -\frac{1}{HW}\sum_{i=1}^{HW} \mathcal{G}_{\overline{\mathcal{Y}}_i^t} \cdot \log P_i^t\left(\overline{\mathcal{Y}}_i^t\right). \tag{16}$$

The dynamic refined loss is utilized as an auxiliary loss to help the model focus more on the under-learning classes during incremental training. These classes are more prone to severe forgetting during multistep training. As shown in the experimental section, this phenomenon is alleviated with our dynamically refined loss.

IV. Experiments and Analysis

A. Datasets and Protocols

Our STCL-DRNet is evaluated on four semantic segmentation datasets in remote sensing scenes: DeepGlobe Land Cover Classification Challenge dataset [19], ISPRS Potsdam and Vaihingen [21], and iSAID [20].

1) DeepGlobe [19]: The DeepGlobe Land Cover Classification Challenge dataset consists of 1146 satellite images, each measuring 2448 × 2448 pixels. These images are divided into three sets: training, validation, and test, with 803, 171, and 172 images, respectively. All images in the dataset include RGB channels, captured with a 50-cm pixel resolution, sourced from the DigitalGlobe Vivid+ collection. Every Satellite image is paired with an annotation mask. These mask categories represent different types of land cover, including water bodies, urban areas, forests, agricultural land, and barren land, which are arranged as follows: 0: unknown, 1: urban, 2: agriculture, 3: rangeland, 4: forest, 5: water, and 6: barren. For training, the dataset provides a set of labeled images, while the test set includes additional satellite images without labels for evaluation purposes. Following [14] and [16], we partition training images into a training set and a test set with 455 images and 348 images.

2) iSAID [20]: The iSAID is a benchmark dataset for segmentation and instance segmentation in aerial imagery from DOTA [53]. This dataset comprises 2806 high-resolution images with 655 451 object instances across 15 categories and background. The categories are arranged as follows: 0: background; 1: baseball diamond; 2: basketball court; 3: bridge; 4: ground track field; 5: harbor; 6: helicopter; 7: large vehicle; 8: plane; 9: roundabout; 10: ship; 11: small vehicle; 12: soccer ball field; 13: storage tank; 14: swimming pool; and 15: tennis court. It consists of 1411 images in the training set, 458 images in the validation set, and 957 images in the test set. Following [16], [17], 20% of the training images are used for validation. The validation set is used for test.

3) ISPRS Potsdam and Vaihingen [21]: The ISPRS 2-D Semantic Labeling contest provides two airborne image datasets: Potsdam and Vaihingen. The Vaihingen 2-D semantic labeling dataset consists of 33 patches, measuring 2494 × 2.064 pixels on average. The Potsdam 2-D dataset

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consists of 38 patches with the same shape of 600 × 6.000 pixels. They are all comprised of very high-resolution true ortho photo (TOP) tiles and corresponding digital surface models (DSMs) derived from dense image matching techniques. Each dataset has been classified manually into six most common land cover classes: 0: background (clutter); 1: building; 2: car; 3: impervious surface; 4: low vegetation; and 5: tree. Following [17], we adopt 16 images defined by the organizer for training and the remaining 17 for testing in the Vaihingen dataset. For the Potsdam dataset, we use 24 out of 38 patches as the training set and the remaining 14 patches as the test set.

4) Metric: Following [14], [16], and [17], we utilize the Pascal VOC mean intersection-over-union (mIoU) to estimate the performance. For the step t, we compute the mIoU for each class in learned classes $C^{0:t}$, as denoted in the following equation:

$$\text{mIoU}^t = \frac{1}{|C^{0:t}|}\sum_{i}^{|C^{0:t}|}\frac{p_{ii}}{\sum_{j=1}^{|C^{0:t}|}p_{ij}+\sum_{j=1}^{|C^{0:t}|}p_{ji}-p_{ii}} \tag{17}$$

where p_{ij} denotes the number of pixels belonging to class i is classified as class j, $|C^{0:t}|$ is the number of classes which is learned from step 0 to step k. Following previous works, we report the final mIoU results at the end of the final step T.

5) Protocols: Following [9] and [10], two incremental learning protocols for semantic segmentation tasks are proposed: **overlapped** and **disjoint**. The shared characteristic of these two protocols is that the annotations for step t images X^t only contain new classes C^t and the background class c_b. For **disjoint** protocols, the ground truth classes only include $C^{0:t}$, which means the pixels with ground truth $C^{0:t-1}$ are labeled as background c_b. But for **overlapped** protocols, the ground truth includes all the classes $C^{0:T}$ from all steps. As a result, the labels for both the previously learned classes $C^{0:t-1}$ and the classes to be learned in the future $C^{t+1:T}$ are all labeled as the background class c_b. By contrast, the **overlaped** protocol is more challenging and difficult for dealing with confusion with more classes. In our research, we focus on the **overlaped** protocol in all our experiments.

B. Implementation Details and Configurations

1) Implementation Details: To ensure a fair comparison between our method and previous works, we use the DeepLabV3 [7] with ResNet101 [55] backbone as our basic segmentation model. The backbone ResNet101 is pretrained on the Imagenet-1k [56]. To ensure robust results, all experimental results are averaged over five independent runs with stochastic random seeds. The observed consistency across these runs underscores the stability and general reliability of the model's performance. In the training of our model, we use a stochastic gradient descent (SGD) optimizer with a weight decay $1e^{-4}$ and a batch size 24. The learning rate is initialized lr = 0.01 in the first step and lr = 0.001 in the subsequent steps, decaying according to a polynomial schedule. For data augmentation, we employ standard augmentation techniques, including random scaling between 0.5 and 2.0, random cropping to a size of 512, and random flipping. Specifically for the iSAID dataset, we crop the images into a size of 512 × 512 pixels with the overlap sliding windows of 384, following [16], [17]. We crop the 16 training patches from Vaihingen and 24 patches from Potsdam into the size of 512 × 512 pixels with the overlap of 128. The hyperparameters are set as momentum factors $\gamma = 0.99$, training stage $S = 5$ ratio, and loss factor $\alpha = 2$. The EMA model $\overline{f}$ and gradient scores $\mathcal{G}_c$ are updated every iteration. Implementation of our work is based on Pytorch 1.8 with CUDA 11.3, and all experiments are conducted on a workstation with 1 NVIDIA 4090 GPU or 1 NVIDIA 3090 GPU. The code is available at https://github.com/cv516Buaa/STCL-DRNet

TABLE I

COMPARISON RESULTS ON DEEPGLOBE DATASET IN MIOU (%) IN THE SETTING OF 3-3 AND 2-2S. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD. THE "JOINT" AND "$x-y[s]$" SETTINGS ARE EXPLAINED IN SECTION IV-B2

Method	3-3 (2 steps)			2-2s (3 steps)			
	1-3	4-6	all	1-2	3-4	5-6	all
FT	0.0	61.3	30.6	0.0	0.0	65.5	21.8
LWF [38]	42.9	40.7	41.8	44.9	33.1	38.5	38.8
LWF-MC [27]	57.2	46.1	51.6	57.4	38.5	43.8	46.6
MiB [9]	53.4	17.0	35.1	57.2	19.9	27.9	35.0
ILT [54]	49.9	45.0	47.4	56.0	34.1	46.7	45.6
Disall [14]	60.7	50.1	55.4	61.2	47.1	51.6	53.2
PSG-TWD [16]	65.6	54.7	60.1	62.7	49.9	65.5	59.4
Ours	**67.8**	**71.4**	**69.6**	**81.2**	**56.7**	**69.4**	**69.1**
Joint	67.8	73.8	70.8	82.2	59.2	71.1	70.8

2) Incremental Configurations: Following [14], [15], [16], [17], we set up multiple incremental experiments for each dataset. We define our incremental settings as "$x-y[s]$," where "x" is the number of classes learned in the first step, "y" is the number of classes learned in subsequent steps, and "s" indicates more than two steps in total. For the DeepGlobe dataset, we evaluate our method on 3-3 (2 steps), 2-2s (2 steps), and 1-1s (6 steps) settings. For example, the 2-2s setting means learning three classes at the first step and learning the remaining classes 2 × 2 in subsequent steps, which totally needs 3 steps for all six classes. For the sake of comparison, we adopt the learning sequence defined in the experimental setup of [14], [16]. As for the iSAID dataset, we conduct the experiments on the settings of 14-1 (2 steps), 10-5 (2 steps), and 5-5s (3 steps). The learning sequence follows [17] in alphabetical order. The Potsdam and Vaihingen datasets share the same classes and class count. We evaluate our method on the same settings of 4-1 (2 steps), 3-1s (3 steps), 2-1s (4 steps), and 1-1s (5 steps). The class learning sequence is also arranged in alphabetical order. To further validate the effectiveness of our method, we introduce the static training results as the upper bound for incremental learning, referred to as "joint." In the "joint" experiments, all data annotated with all classes are provided simultaneously.

C. Quantitative Evaluation

1) Class-Incremental Learning in DeepGlobe: We first evaluate our proposed method on the class-incremental semantic segmentation task in DeepGlobe. Following the overlapped protocol in Section IV-B2 and Table I present the comparition

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TABLE II
COMPARISON RESULTS ON DEEPGLOBE DATASET IN MIOU (%) IN THE SETTING OF 1-1S. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD. THE SETTINGS ARE EXPLAINED IN SECTION IV-B2

Method	1	2	3	4	5	6	all
	Urban	Agriculture	Rangeland	Forest	Water	Barren	all
FT	0.0	0.0	0.0	0.0	0.0	86.7	14.4
LWF [38]	40.2	44.4	2.0	29.4	45.7	28.5	31.7
LWF-MC [27]	32.9	56.8	11.5	39.3	72.2	44.4	42.8
MiB [9]	52.1	50.4	20.2	20.4	34.7	33.9	35.3
ILT [54]	35.5	57.8	16.6	35.9	65.0	49.0	43.3
Disall [14]	52.4	62.1	29.9	42.2	73.1	55.1	52.4
PSG-TWD [16]	54.7	63.5	**40.7**	42.6	74.2	66.4	57.0
Ours	**78.3**	**83.4**	32.6	**76.8**	**75.4**	**64.2**	**68.6**
Joint	77.1	86.8	40.2	78.2	77.9	64.3	70.8

of our method with existing works [9], [14], [16], [27], [38], [54] across three experimental settings("3-3", "2-2s", and "1-1s"), showing our superiority of our work. In all settings, our method achieves the best results which is highlighted in bold. The mIoU results after all step training, including those from the initial step (step 0) and the subsequent incremental steps, are listed.

In the two-step setting "3-3," our STCL-DRNet outperforms the existing SOTA method PSG-TWD [16] by 9%–10% in terms of mIoU. As shown in Table I, directly fine-tuning (FT) leads to significant catastrophic forgetting of old classes from the previous step. Most methods achieve excellent segmentation accuracy for old classes in the first step but perform poorly on new classes in the second step, with performance inferior to our method by at least 17%. Besides, in the three-step setting "2-2," our method STCL-DRNet still performs better than the existing SOTA method by 9%–10%. Notably, for the two old classes in the first step, our proposed STCL-DRNet surpasses the SOTA method by nearly 20%, demonstrating the effectiveness of our method in mitigating catastrophic forgetting of old classes. To further validate the effectiveness of our method, we conducted experiments in the most challenging "1-1" setting. As compared in Table II, our STCL-DRNet is superior to the existing SOTA methods by 11% in terms of mIoU. Especially for the old classes *urban*, *agriculture*, and *forest*, our method STCL-DRNet achieves the best results, surpassing the existing SOTA method PSG-TWD by 20%–30%. "But for rare classes such as *Rangeland*, our approach is only slightly inferior to PSG-TWD, which suggests that generative methods like PSG-TWD may be more effective in handling difficult or rare classes. However, our method achieves more balanced results across all classes, as evidenced by the highest result of mIoU.

2) Class-Incremental Learning in iSAID: We further conduct experiments on the iSAID [20] dataset, and compare our method with SOTA methods in three settings ("10-5," "5-5s," and "10-1s"). As compared in Table III, our STCL-DRNet outperforms previous methods in all experiments, including "10-5," "5-5s," and "10-1s" in terms of mIoU. It is evident that our proposed method performs significantly better when handling longer incremental steps. In the two-step setting "10-5," our method STCL-DRNet outperforms the previous SOTA method Micro [17] by only 2% in terms of mIoU, and is slightly inferior on the five classes of the second step. But for incremental learning with longer steps, our STCL-DRNet performs much better than previous methods both in learning new-class knowledge and retaining old-class knowledge. In the "5-5s" setting, our method STCL-DRNet outperforms the existing method by 4%, especially on the old classes in the first step, where it surpasses the SOTA method by nearly 15%. The performance is more pronounced in the five-step setting "10-1s." Our method STCL-DRNet achieves excellent performance with 44.64% mIoU, surpassing existing methods by at least 10%, which reflects the effectiveness of our approach in reducing class confusion caused by errors in multistep learning.

3) Class-Incremental Learning in ISPRS Potsdam and Vaihingen: To further verify the robustness of our proposed method, we conduct experiments on the Potsdam and Vaihingen datasets and compare it with SOTA methods across three scenarios. The results for Potsdam and Vaihingen are presented in Tables IV and V. In the Potsdam dataset, our method STCL-DRNet achieves stable mIoU results of 83% across all scenarios (3-1, 2-1s, and 1s), demonstrating the robustness of our approach in handling incremental training with different step configurations. In the Vaihingen dataset, our method STCL-DRNet outperforms existing methods in all scenarios by 5%–7% in terms of mIoU. Furthermore, our method achieves significant improvements not only in the average IoU but also in both new and old classes during the staged learning process. Our method surpasses other existing methods by 7%–10% in IoU for both old and new classes across all settings.

D. Examples Visualization and Analysis

To better illustrate the performance of our proposed method, we present the segmentation masks predicted by the model and ground truth from different validation image set in Figs. 5–7. Figs. 5 and 6 are inference examples from DeepGlobe dataset. Our proposed method achieves similarly good results across different lengths of incremental experimental settings. Even in the most challenging 1-1 setting, our method still achieves good results, despite some minor confusion in the details. Fig. 7 shows the visualization results in the iSAID dataset which also demonstrates the robust performance across different incremental settings.

E. Ablation Study

To further analyze our method STCL-DRNet, we conduct a detailed ablation study in the incremental setting "5-5s" of the iSAID dataset. Since some parts of our proposed method are interdependent, we add the components sequentially—namely, EMA branch, dynamic CL, and dynamic refined loss—to evaluate their individual contributions.

Table VI shows the results of our ablation study on sequentially adding components in the "5-5s" scenario of the iSAID dataset. The baseline model is trained using only the supervision loss L_{ce} from the frozen model f_{t-1} of the last step. As indicated in Table VI, adding the EMA branch improves the ability to learn new classes in subsequent steps, but slightly reduces the retention of knowledge for old classes. Introducing dynamic refined loss L_{dr} in the EMA branch,

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TABLE III
COMPARISON RESULTS ON ISAID DATASET IN MIOU (%). THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Method	10-5 (2 steps)			5-5s (3 steps)				10-1s (6 steps)		
	1-10	11-15	all	1-5	6-10	11-15	all	1-10	11-15	all
FT	0.00	51.10	17.03	0.00	0.00	47.64	15.88	0.00	15.63	5.21
LWF [38]	44.70	45.41	44.94	36.53	26.70	38.44	33.89	4.99	16.81	8.93
ILT [54]	49.25	35.14	44.55	40.80	9.68	31.30	27.26	15.58	19.46	16.87
MiB [9]	47.80	47.34	47.65	46.75	17.83	37.54	34.04	25.06	22.94	24.35
HCISS [15]	52.52	39.52	48.19	43.54	23.94	35.67	34.39	31.48	27.65	30.21
PSG-TWD [16]	53.81	40.22	49.28	44.60	25.89	34.73	35.07	34.35	26.43	31.71
Micro [17]	49.17	**56.90**	51.75	44.09	**43.22**	50.28	45.86	28.82	38.01	31.88
ours	**54.25**	53.33	**53.94**	**59.62**	34.03	**53.62**	**49.09**	**46.57**	**41.71**	**44.64**
Joint	57.97	61.92	59.29	70.69	45.25	61.92	59.29	57.97	61.92	59.29

TABLE IV
COMPARISON RESULTS ON POTSDAM DATASET IN MIOU (%). THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Method	3-1 (2 steps)			2-1s (3 steps)			1s (5 steps)		
	1-3	4-5	all	1-2	3-5	all	1	2-5	all
FT	0.00	35.54	14.22	0.00	22.94	13.76	0.00	16.80	13.44
MiB [9]	83.42	70.28	78.17	83.08	70.06	75.27	88.49	67.83	71.96
HCISS [15]	83.56	70.82	78.47	83.54	70.11	75.49	89.28	67.22	71.64
Micro [17]	84.03	70.88	78.77	83.14	72.57	76.80	89.43	70.69	74.44
ours	**89.15**	**76.04**	**83.91**	**90.34**	**79.15**	**83.63**	**92.73**	**80.85**	**83.23**
Joint	89.62	76.85	84.51	91.26	80.01	84.51	93.17	82.35	84.51

TABLE V
COMPARISON RESULTS ON VAIHINGEN DATASET IN MIOU (%). THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Method	3-1s (2 steps)			2-1s (3 steps)			1s (5 steps)		
	1-3	4-5	all	1-2	3-5	all	1	2-5	all
FT	0.00	36.87	14.75	0.00	24.85	14.91	0.00	18.44	14.75
MiB [9]	74.30	68.23	71.87	70.76	70.09	70.36	86.01	50.93	57.95
HCISS [15]	74.19	68.53	71.93	71.28	69.61	70.28	**87.03**	50.93	58.15
Micro [17]	74.77	67.75	71.96	69.59	70.97	70.42	86.84	56.42	62.54
ours	**78.67**	**73.38**	**76.55**	**76.87**	**77.19**	**77.06**	76.76	**65.90**	**68.07**
Joint	80.52	75.21	78.40	78.38	78.41	78.40	90.03	75.49	78.40

TABLE VI
ABLATION STUDY ON ISAID 5S IN MIOU (%)

Method	1-5	5-10	10-15	all
Baseline	**61.06** ± 0.82	30.58 ± 0.61	51.50 ± 1.12	47.69 ± 0.85
+EMA Branch	59.00 ± 0.52	31.15 ± 0.73	53.32 ± 0.94	47.82 ± 0.73
+Curriculum Learning	59.21 ± 0.71	31.85 ± 0.56	**53.68** ± 0.43	48.24 ± 0.57
+Dynamic Refined Loss	59.62 ± 0.76	**34.03** ± 0.63	53.62 ± 0.47	**49.09** ± 0.62

TABLE VII
MODULE CONTRIBUTION STUDY ON ISAID 5S IN MIOU (%)

Last Step Model f_{t-1}		EMA Branch $\overline{f}$			mIoU
L_{ce}	CL	L_{ce}	L_{dr}	CL	
✓					47.69 ± 0.85
✓	✓				43.58 ± 0.61
✓		✓			47.82 ± 0.73
✓		✓		✓	48.24 ± 0.57
✓			✓	✓	**49.09** ± 0.62

which assigns weights to under-learning classes, significantly improves performance across all classes, especially for those with low performance. The significant improvement achieved by L_{dr} is partly due to the inherent class imbalance in the iSAID dataset, which is a common phenomenon in remote sensing imagery.

To further reveal the contribution of each module in the proposed method, we disclose some modules individually or swap them with common alternatives, such as removing the

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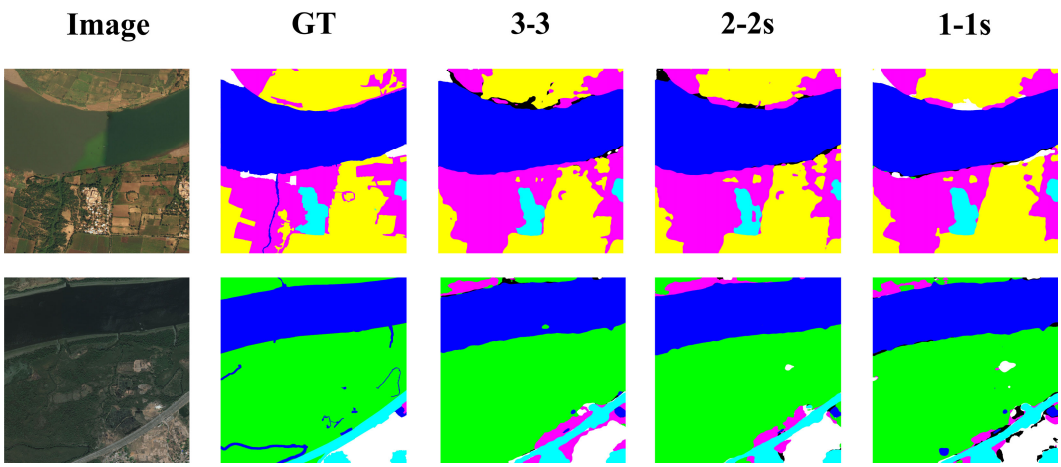


Fig. 5. Visualization results of our method in DeepGlobe 3-3, 2-2s, and 1-1s scenarios.

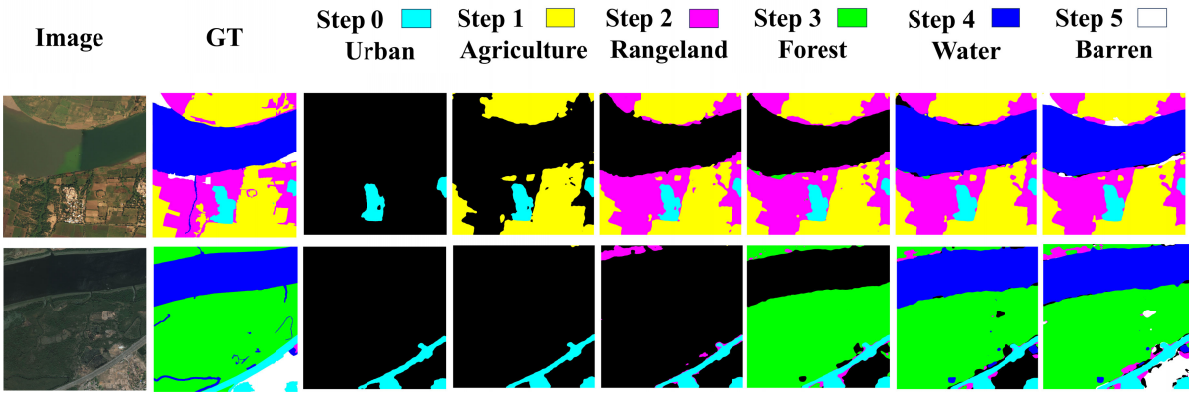


Fig. 6. Visualization results inferred by our model trained after each step on the DeepGlobe 1-1 scenario.

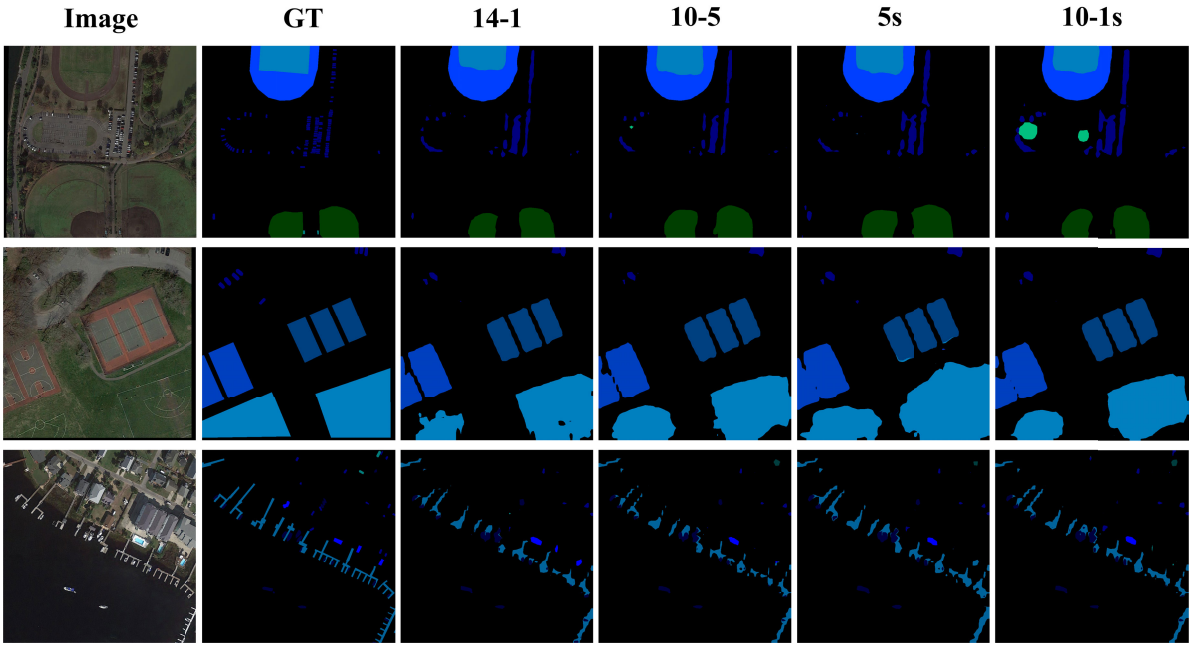


Fig. 7. Visualization results of our method in iSAID 14-1, 10-5, 5s, and 10-1s scenarios.

EMA branch and replacing L_{dr} with L_{ce}. As shown explicitly in Table VII, without the EMA self-training branch, directly applying CL to the pseudo-label threshold selection of the previous step model leads to a decrease in overall training per-

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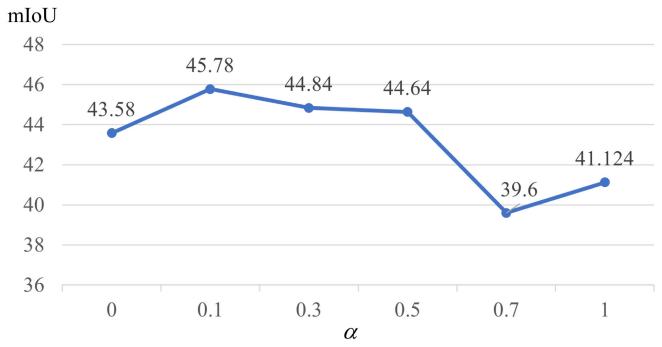


Fig. 8. Final mIoU(%) results on iSAID 10-1s under the effect of the loss parameter α.

formance. This can be interpreted as not exposing the "hard" labels to the model being detrimental to the training process, as it prevents the model from learning the finer details of the segmentation task. The boundary regions between different class segmentation masks often have lower softmax scores, and learning from these regions is crucial for distinguishing between classes. Therefore, in the following experiments, we introduce CL only to the EMA branch, allowing the model to retain as much detailed information as possible from the last-step model for old classes. Without CL, the improvement from introducing only the self-training branch and using cross-entropy loss L_{ce} is relatively small. Combining CL with the self-training branch yields a 1% improvement in overall performance. Replacing L_{ce} with dynamic refined loss L_{dr} on the EMA branch $\overline{f}$ boosts mIoU by nearly 1%, primarily enhancing performance on weak classes. The resulting standard deviations demonstrate that, with the integration of CL and the EMA self-training branch, our method also maintains good stability. Overall, each module in our method is effective and appropriately combined, leading to the best results.

F. Impact of Hyperparameters

We analyze the impact of two key hyperparameters: the weight of dynamic refined loss α in Section III-B and the interval of EMA updating ζ in Section III-C. Experiments are conducted in the 10-1s overlap setting on the iSAID dataset. Unless otherwise stated, other parameters are theoretically fixed (e.g., $\gamma_m = \gamma_i = 0.99$). We compare the final mIoU results to reveal the Impact of hyperparameters. Fig. 8 shows the effect of α in the challenging 10-1s setting. The model achieves significant improvements when α lies between 0.1 and 0.5, while performance degrades as α exceeds 0.7. Our method shows consistent performance for $\alpha \in [0.1, 0.5]$, indicating moderate sensitivity to this parameter within a reasonable range. A very small α may underutilize the dynamic refinement, while an overly large α could destabilize the main training objective. As summarized in Table VIII, we also evaluate the influence of the EMA update interval ζ. The best performance is attained when the EMA teacher is updated after every iteration ($\zeta = 1$). Longer intervals (e.g., $\zeta > 1$) lead to performance drops, suggesting that frequent EMA updates are essential for maintaining teacher stability. Based on these findings, we set $\alpha = 0.5$ and $\zeta = 1$ for all datasets and incremental scenarios.

TABLE VIII

FINAL MIOU(%) RESULTS ON ISAID 10-1S UNDER THE EFFECT OF UPDATING INTERVALS ζ FOR THE EMA MODEL

Intervals ζ	1 iteration	10 iterations	100 iterations	1 epoch
mIoU	**44.64**	43.38	41.25	38.61

TABLE IX

COMPUTATIONAL OVERHEAD OF THE EMA UPDATE AND DUAL-TEACHER FRAMEWORK ON A SINGLE RTX 4090

Dataset	DeepGlobe	iSAID
Image Resolution	8	32
Batch Size	1024	512
Time per Iteration (ms)	470	490
Time per EMA Update (ms)	21	27
Two-teacher Inference Time (ms)	60	68

TABLE X

MEMORY FOOTPRINT CONTRAST BETWEEN ONE-TEACHER METHOD AND DUAL-TEACHER METHOD (OURS)

Method	One-teacher Method(MiB [9])	Two-teacher Method(Ours)
Backbone Params (M)	59.81	59.81
Backbone Memory (MB)	230.89	230.89
Teacher Memory (MB)	227.13	458.27
Peak Training Memory (MB)	20773.05	21734.32

G. Limitations and Future Works

1) Limitations: Although the proposed method yields effective results, several limitations warrant further investigation. First, during training, our method requires two-teacher models to perform inference simultaneously, which inevitably incurs higher GPU memory overhead than distillation approaches that use a single teacher model. Table IX examines the computational overhead introduced by the EMA update and the dual-teacher setup. Although the EMA branch adds computational overhead, it represents a minor portion of the iteration time, which is dominated by the model's backward pass and gradient update. Table X compares the memory footprint of single-teacher methods (e.g., MiB) and our dual-teacher method. The dual-teacher model's memory overhead scales with model size, while in practice, it was minimal compared to the data memory. Our method adds only minimal latency, making it applicable to real-time scenarios. However, its computational and memory costs grow faster with the size of the dual-teacher model than the one-teacher method. Consequently, training with large-scale models requires substantial resources.

Second, we attribute catastrophic forgetting of old classes to the distribution gap between new data at the current step and old data from earlier steps. Quantifying this visual-domain discrepancy is challenging, as no single metric fully captures its complexity.

Third, the integration of an EMA teacher with CL mitigates model degradation by reducing output errors from the teacher.

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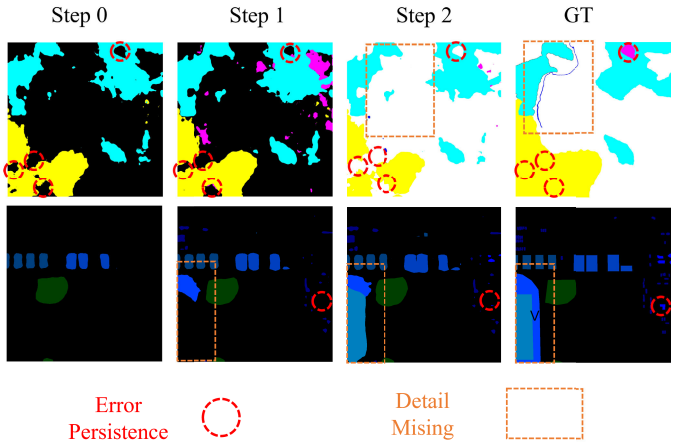


Fig. 9. Failure case illustrations in the iSAID 5-5 (3-step) and DeepGlobe 2-2 (3-step) settings.

However, the overall performance is still ultimately bounded by the quality of the historical knowledge. As shown in Fig. 9, some persistent challenges remain for our method in class-incremental segmentation. Errors acquired in the early training stage are prone to propagation throughout subsequent steps, making them particularly persistent and challenging to rectify. Another key challenge in class-incremental learning is modeling the semantic, spatial, and structural relationships among classes introduced incrementally. This leads to a loss of detail in the segmentation output.

2) Future Works: In future work, we plan to expand our research in several promising directions. First, beyond the class-incremental learning, we will explore more incremental learning paradigms, such as domain-incremental learning and modality incremental learning. They are closely related to current multimodal fusion and cross-domain image representation, as discussed in [57] and [58]. Second, we will explore latent features and graph neural networks in class-incremental learning, as introduced in [59], to model complex relationships between classes in different class-incremental learning steps. Third, multitype semi-supervised learning provides a means to cope with limited data for certain old classes, as highlighted in a self-paced manner in [52]. These could be adapted to improve the class-incremental segmentation accuracy for future directions.

V. Conclusion

In this article, we propose an incremental semantic segmentation method in remote sensing scenes STCL-DRNet. Apart from catastrophic forgetting and background drift, the other two challenges inherent in remote sensing data that could exacerbate the above-mentioned issues are also highlighted, which are uneven changing class distribution and error accumulation-induced degradation. To mitigate these issues, we propose a self-training auxiliary branch to integrate cross-step model knowledge and prevent rapid forgetting in the incremental training process. In addition, by integrating a class-balanced curriculum training strategy with the self-training branch and employing a dynamic, refined loss, our approach alleviates performance degradation resulting from noisy accumulation. Extensive experiments demonstrate the superiority of STCL-DRNet over SOTA methods. Visual analyses further validate the effectiveness. Our future work will focus on better exploring the intrinsic relationship between incremental training and static training objectives, with the goal of more effectively addressing the degradation problem in incremental learning.

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